

streamlit app

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BTC Sentiment-Driven LSTM Trading Dashboard

An end-to-end ML pipeline combining FinBERT news sentiment with LSTM price forecasting, Optuna hyperparameter optimization, Kelly-based risk management, and SHAP interpretability.

🔴 Live BTC-USD Price

BTC-USD

24h Change

\$59,2...

-1.19%

↓ -1.19%



📅 Data via yfinance • Cached for 5 minutes • Last updated: 2026-06-30 09:55 UTC



Live Crypto News Sentiment

Latest Headline:

"Grayscale CEO Calls for Simultaneous Approval of Spot Products to Level the Field..."

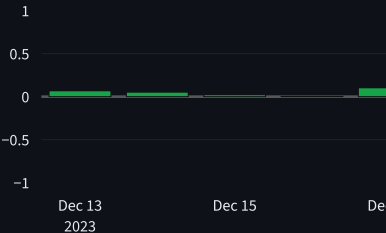
📅 Source date: 2023-12-19 06:40:41

Sentiment Score

-0.100

🟡 Neutral

7-Day Sentiment Trend



📄 Data from cryptonews.csv (GitHub raw) • TextBlob sentiment • Cached for 10 minutes



Key Performance Metrics

Best LSTM Sharpe (ba...

3.11

↑ 1.6472

Buy & Hold Sharpe

2.96

↑ 1.6156

Best LSTM Max Drawd...

-10.79%

↑ 1.93pp vs B&H

Optuna Best OOF Sha...

2.33

↑ 4 trials

Risk-Managed Max DD

-0.37%

↑ 21 trades



Strategy Comparison — Phase 1 (LSTM Configs vs Buy & Hold)

Strategy	Final Portfolio Value	Sharpe Ratio	Sortino Ratio	Max Drawdown	Win %
baseline	1.6472	3.1071	5.9353	-10.79%	53.64%
lightweight	1.0439	0.4759	0.6191	-16.87%	23.64%
regularized	1	0	0	0.0%	0.0%
bilstm	1.2052	2.1804	3.7299	-4.06%	12.73%
buy_hold	1.6156	2.9633	5.6554	-12.72%	54.55%



Phase 2: Risk-Managed Backtest Comparison

strategy	final_value	sharpe	max_dd	n_trades	circuit_breaker	Risk-Managed Final Value
Simple (all-in/out)	1.5658	2.8439	-0.1206	1	☐	1.0154
Risk-Managed (Kelly+Vol+DD)	1.0154	2.6369	-0.0037	21	☐	
Buy & Hold	1.6156	2.9633	-0.1272	1	☐	2.6369

Circuit Breaker Triggered

No



Optuna Best Hyperparameters

<pre>{ "Learning Rate" : "0.000561" "LSTM Units" : 32 "Dropout" : 0 "Number of Layers" : 1 }</pre>	<pre>{ "Best OOF Sharpe" : "2.3299" "Total Trials" : 5 "Completed Trials" : 4 "Pruned Trials" : 1 "Walk-Forward Folds" : 5 }</pre>
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Navigation: Use the sidebar to explore **Phase 1** (data → sentiment → LSTM → backtest) and **Phase 2** (walk-forward CV → Optuna → risk-managed backtest → SHAP) deep-dives.